

Rajdeep Gill

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Education

University of Manitoba

Bachelor of Science in Computer Engineering

Expected May 2026

4.46/4.5 GPA

Skills

Programming Languages: TypeScript, JavaScript, Python, C++, Java

Frameworks and Libraries: React, NextJS, PyTorch, OpenMPI, Node.js, Express, Flask

Tools: Git, Docker, PostgreSQL, Github Actions, Bazel, gRPC, Protocol Buffers (Protobuf)

Experience

Robinhood | Software Engineer Contractor

Sep 2025 - May 2026

- Redesigned the margin call interface with intuitive risk indicators, detailed requirement breakdowns, and actionable recovery steps to help users better understand their margin obligations and reduce overall fund recovery time
- Built tax lot selling feature using observables for real-time market data updates and implemented virtualization to ensure performant rendering for users with thousands of tax lots

Robinhood | Software Engineer Intern

May 2025 - Sept 2025

- Developed and shipped a real-time margin maintenance table built with React and Typescript, featuring embedded trade execution and risk indicators.
- Overhauled the margin experience by expanding the server-driven UI to include futures traders and unifying business logic across Web, iOS, and Android, resulting in a 43% increase in margin usage.
- Enhanced a Protobuf-based server-driven UI architecture by implementing 10+ foundational components in Python and Typescript along with a logging framework to increase system usage and maintainability
- Repaired the core integration suite for the margin experience and added 20+ test suites, achieving 93% code coverage

Biosensors Research Lab | Machine Learning Research Intern

May 2024 - Dec 2024

- Created a custom convolutional neural network to predict cell viability from optical microscopy images, achieving 85% accuracy, establishing a non-invasive alternative to traditional staining methods for assessing cell health
- Developed a model validation GUI in Python displaying SHAP-based feature attributions, and visualizing cell images to debug model predictions
- Created a parallelized data processing pipeline with a new morphological parameter extraction algorithm, improving accuracy by 17% and cutting processing runtime by 75%.
- Replaced a high-risk system of manually-shared hard drives with a centralized MySQL database, allowing for cross-experiment queries and easy data sharing

Biosensors Research Lab | Machine Learning Research Intern

May 2023 - Aug 2023

- Built a framework for electronic testing equipment, allowing for automation of entire experimental sequences through Python scripts and cutting manual setup time
- Programmed an FPGA to coordinate microfluidic device operations, integrating di-electrophoresis control with synchronized LED illumination and cell imaging

Projects

Void - AI App Builder (Next.js, TypeScript, Prisma, Docker)

- Developed an app to generate and deploy web apps from user prompts, integrating a usage-based Stripe billing system to monetize user compute time and API token consumption
- Implemented a persistent background job queue to process long-running AI build tasks asynchronously, allowing users to submit jobs and receive the completed result at a later time, even after closing the browser
- Utilized Docker for user code sandboxing, secure code execution and real-time application previews

Bug Trace - Developer Workflow Tracker (Python, FastAPI, LangChain, Docker)

- Built a Dockerized backend service to track and evaluate developer debugging workflows by capturing file operations, terminal commands, test results, and pull requests
- Implemented automated test suite execution and validation to verify bug fixes and detect breaking changes, ensuring only successful debugging sessions contributed to the training dataset
- Generated high-quality training data from successful debugging sessions to fine-tune code generation models